MIT SCHOOL OF COMPUTING
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A Leap Towards World Class Education

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Subject: IDS Roll No.: 07 Exp. No.: 2
Name of the Experiment:
Performed on:
Submitted on:

Marks	Teacher's Signature with date
10	

IDS Assignment - 2

1. Sequential organization & characteristics of linear data structures.

it is a method of starting data element in a continuos memory location where elements ~~in a~~ are arranged one after another in a fixed order.

characteristics

- elements are stared in contiguos memory location
- each elements has a unique index or position
- ~~Dis~~ Direct is arranged in a linear sequence.
- size is usually fixed.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

2. Array as an abstract Data types (ADT)

ADT is collection of elements of the same data type stared in contiguous memory location & accessed using an index.

- Component of ADT :

→ Elements : Data items of the same type
→ index : position used to access elements
→ size : Total number of elements

- Basic operations on array.

→ Transversal : visiting each elements
→ insertion : Adding a elements
→ Deletion : Removing an elements
→ searching : Finding an elements

3. insertion, deletion & searching in a 1D array.

consider an array :

A = [10, 20, 30, 40]

insertion : insert is a position 2 :

Shift element right → A = [10, 20, 25, 30, 40]

- Deletion : Delete element at position 3 :

  shift elements left → A = [10, 20, 25, 40]

- Searching : To search 25

  Compare sequentially until found

  25 is found in index 2.

4. Advantages & Limitations of arrays.

Advantages :-

- Fast direct access using index (O(1))
- simple & easy to implement
- effeciently memory usage for fixed size data
- good for staring homogenous data

Limitations :

- Fixed size
- insertion & detection are time - constant
- wastage of memory if size not useful.

efficiently comparision:

- insertion: O(n) due to shifting
- Deletion: O(n) due to shifting
- searching: O(n) for linear search
  O(logn) for binary search (sorted array)

5. 2-Dimensional array for student academic data:

2D array: Rows represent student
columns represents marks/subject.

Student	Maths	Science	English
S_1	85	90	88
S_2	78	82	80

Accessing data:

marks [0][1] → Science marks
Transversal uses nested loops

Efficient manpuilation
- Direct indexing allows fast access
- Nested loop helps is processing Total & Averges

Extension to n-dimensional arrays:

- image processing
- scientific simulations
- ML & data analysis
- game developments.